

Website Performance analysis project

Load the libraries and dataset

```
In [51]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df=pd.read_csv(r'C:\Users\abrar\Downloads\data-export (1).csv')
df.head()
```

```
Out[51]: # -----
-----
----- Unnamed: 1 Unnamed: 2 Unnamed: 3 Unnamed: 4 Unnamed: 5 Unnamed: 6 Unnamed: 7
-----
---
```

	Session primary channel group (Default channel...	Date + hour (YYYYMMDDHH)	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session
0	Direct	2024041623	237	300	144	47.526666666666700	0.6075949367088610	4.673333333333330
2	Organic Social	2024041719	208	267	132	32.09737827715360	0.6346153846153850	4.295880149812730
3	Direct	2024041723	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030
4	Organic Social	2024041718	187	256	125	32.16015625	0.6684491978609630	4.078125

Basic Data Cleaning

Fixed the real columns name of the dataset

```
In [52]: df.columns=df.iloc[0]
df.head()
```

Out[52]:

	Session primary channel group (Default channel group)	Date + hour (YYYYMMDDHH)	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session	Enga
0	Session primary channel group (Default channel...	Date + hour (YYYYMMDDHH)	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session	Eng
1	Direct	2024041623	237	300	144	47.526666666666700	0.6075949367088610	4.673333333333330	
2	Organic Social	2024041719	208	267	132	32.09737827715360	0.6346153846153850	4.295880149812730	0.49438
3	Direct	2024041723	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030	0.493562
4	Organic Social	2024041718	187	256	125	32.16015625	0.6684491978609630	4.078125	



Remove the repeated row from the dataset

```
In [53]: df=df.drop(index=0).reset_index(drop=True)
df
```

Out[53]:

	Session primary channel group (Default channel group)	Date + hour (YYYYMMDDHH)	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session	
0	Direct	2024041623	237	300	144	47.526666666666700	0.6075949367088610	4.673333333333330	
1	Organic Social	2024041719	208	267	132	32.09737827715360	0.6346153846153850	4.295880149812730	0.
2	Direct	2024041723	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030	0.4
3	Organic Social	2024041718	187	256	125	32.16015625	0.6684491978609630	4.078125	
4	Organic Social	2024041720	175	221	112	46.918552036199100	0.64	4.529411764705880	0.
...	...	...	...	...	...	...	...	...	...
3177	Unassigned	2024042806	0	1	0	0	0	2	
3178	Unassigned	2024043005	0	1	0	0	0	2	
3179	Unassigned	2024043006	0	1	0	0	0	2	
3180	Unassigned	2024050105	0	1	0	0	0	2	
3181	Unassigned	2024050307	0	1	0	0	0	2	

3182 rows × 10 columns



Change the columns name of the dataset

```
In [54]: df=df.rename(columns={
    'Session primary channel group (Default channel group)': 'Channel group',
    'Date + hour (YYYYMMDDHH)': 'DateHour'
```

```
})  
df.head()
```

Out[54]:

	Channel group	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session	Engagemer
0	Direct	2024041623	237	300	144	47.526666666666700	0.6075949367088610	4.673333333333330	
1	Organic Social	2024041719	208	267	132	32.09737827715360	0.6346153846153850	4.295880149812730	0.49438202247
2	Direct	2024041723	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030	0.493562231759
3	Organic Social	2024041718	187	256	125	32.16015625	0.6684491978609630	4.078125	0.488
4	Organic Social	2024041720	175	221	112	46.918552036199100	0.64	4.529411764705880	0.50678733031

**Check the information of the dataset**In [55]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3182 entries, 0 to 3181
Data columns (total 10 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Channel group                        3182 non-null   object
 1   DateHour                            3182 non-null   object
 2   Users                               3182 non-null   object
 3   Sessions                            3182 non-null   object
 4   Engaged sessions                    3182 non-null   object
 5   Average engagement time per session 3182 non-null   object
 6   Engaged sessions per user           3182 non-null   object
 7   Events per session                  3182 non-null   object
 8   Engagement rate                     3182 non-null   object
 9   Event count                         3182 non-null   object
dtypes: object(10)
memory usage: 248.7+ KB
```

There have no null values in the dataset

Change the dateHour column to date time formate

```
In [56]: df.columns
```

```
Out[56]: Index(['Channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions',
               'Average engagement time per session', 'Engaged sessions per user',
               'Events per session', 'Engagement rate', 'Event count'],
              dtype='object', name=0)
```

```
In [57]: df['DateHour']=pd.to_datetime(df['DateHour'],format='%Y%m%d%H',errors='coerce')
df.head()
```

Out[57]:

	Channel group	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session	Engagement
0	Direct	2024-04-16 23:00:00	237	300	144	47.526666666666700	0.6075949367088610	4.673333333333330	
1	Organic Social	2024-04-17 19:00:00	208	267	132	32.09737827715360	0.6346153846153850	4.295880149812730	0.4943820224719
2	Direct	2024-04-17 23:00:00	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030	0.4935622317596
3	Organic Social	2024-04-17 18:00:00	187	256	125	32.16015625	0.6684491978609630	4.078125	0.4882
4	Organic Social	2024-04-17 20:00:00	175	221	112	46.918552036199100	0.64	4.529411764705880	0.506787330316



Convert the data object type to numeric type

```
In [58]: numeric_cols=df.columns.drop(['Channel group', 'DateHour'])
df[numeric_cols]=df[numeric_cols].apply(pd.to_numeric,errors='coerce')
df.head()
```

Out[58]:

	Channel group	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session	Engagement rate	Event count
0	Direct	2024-04-16 23:00:00	237	300	144	47.526667	0.607595	4.673333	0.480000	1402
1	Organic Social	2024-04-17 19:00:00	208	267	132	32.097378	0.634615	4.295880	0.494382	1147
2	Direct	2024-04-17 23:00:00	188	233	115	39.939914	0.611702	4.587983	0.493562	1069
3	Organic Social	2024-04-17 18:00:00	187	256	125	32.160156	0.668449	4.078125	0.488281	1044
4	Organic Social	2024-04-17 20:00:00	175	221	112	46.918552	0.640000	4.529412	0.506787	1001

In [59]: df.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3182 entries, 0 to 3181
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Channel group                        3182 non-null   object
1   DateHour                             3182 non-null   datetime64[ns]
2   Users                                3182 non-null   int64
3   Sessions                             3182 non-null   int64
4   Engaged sessions                     3182 non-null   int64
5   Average engagement time per session  3182 non-null   float64
6   Engaged sessions per user            3182 non-null   float64
7   Events per session                   3182 non-null   float64
8   Engagement rate                      3182 non-null   float64
9   Event count                          3182 non-null   int64
dtypes: datetime64[ns](1), float64(4), int64(4), object(1)
memory usage: 248.7+ KB

```

Creata new columns of Hour extracted from DateHour column

```
In [60]: df['Hour']=df['DateHour'].dt.hour  
df.head()
```

Out[60]:

	Channel group	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session	Engagement rate	Event count	Hour
0	Direct	2024-04-16 23:00:00	237	300	144	47.526667	0.607595	4.673333	0.480000	1402	23
1	Organic Social	2024-04-17 19:00:00	208	267	132	32.097378	0.634615	4.295880	0.494382	1147	19
2	Direct	2024-04-17 23:00:00	188	233	115	39.939914	0.611702	4.587983	0.493562	1069	23
3	Organic Social	2024-04-17 18:00:00	187	256	125	32.160156	0.668449	4.078125	0.488281	1044	18
4	Organic Social	2024-04-17 20:00:00	175	221	112	46.918552	0.640000	4.529412	0.506787	1001	20

Description of the dataset

```
In [61]: df.describe()
```


Out[61]:

	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session	Engagement rate	Event count
count	3182	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000
mean	2024-04-20 01:17:07.278441216	41.935889	51.192646	28.325581	66.644581	0.606450	4.675969	0.503396	242.278441
min	2024-04-06 00:00:00	0.000000	1.000000	0.000000	0.000000	0.000000	1.000000	0.000000	1.000000
25%	2024-04-13 02:15:00	20.000000	24.000000	13.000000	32.103034	0.561404	3.750000	0.442902	103.000000
50%	2024-04-20 02:00:00	42.000000	51.000000	27.000000	49.020202	0.666667	4.410256	0.545455	226.000000
75%	2024-04-26 22:00:00	60.000000	71.000000	41.000000	71.487069	0.750000	5.217690	0.633333	339.000000
max	2024-05-03 23:00:00	237.000000	300.000000	144.000000	4525.000000	2.000000	56.000000	1.000000	1402.000000
std	NaN	29.582258	36.919962	20.650569	127.200659	0.264023	2.795228	0.228206	184.440000

Exploratory Data Analysis

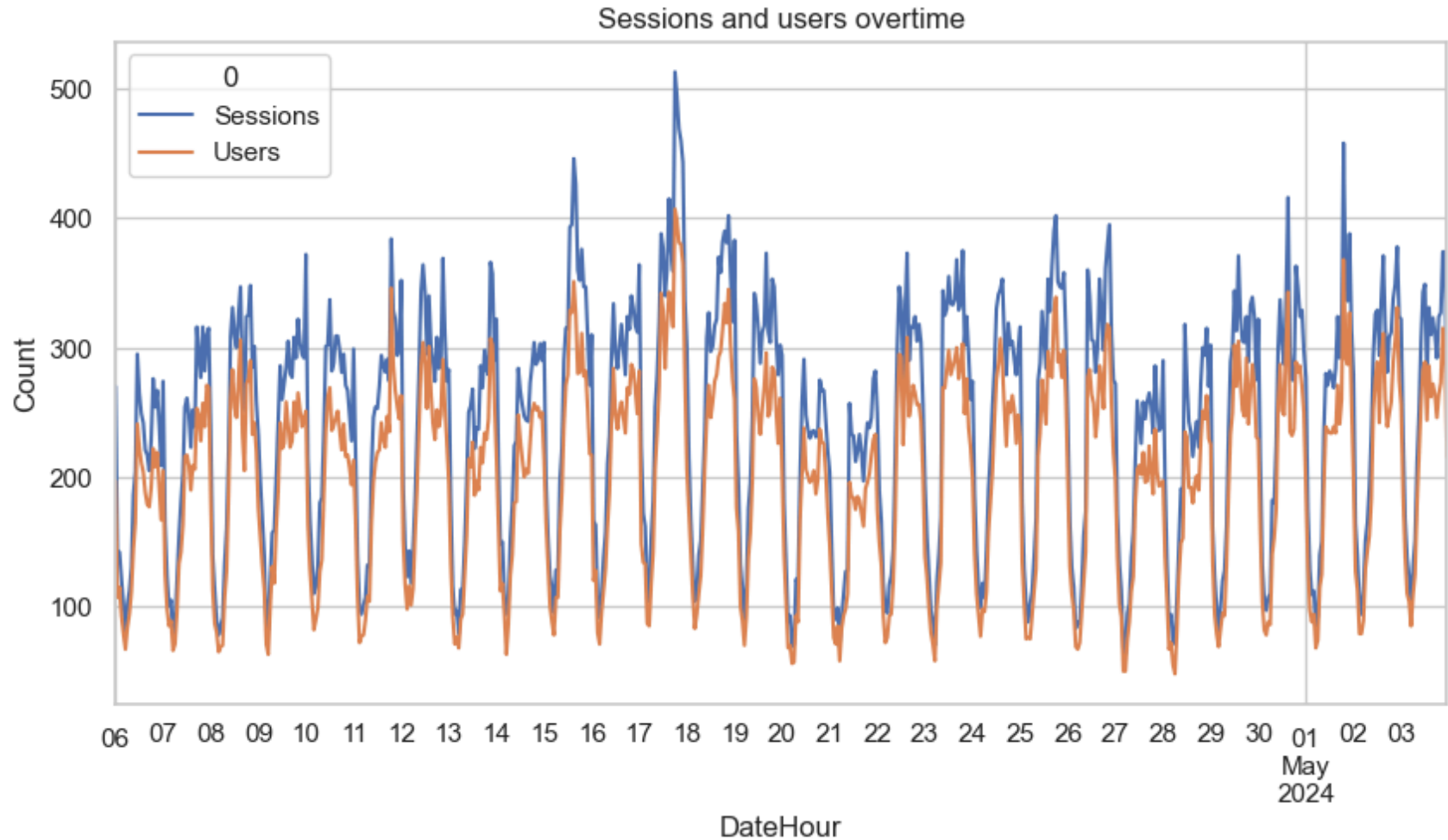
Sessions and users over time

In [62]: `df.columns`

Out[62]: Index(['Channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions', 'Average engagement time per session', 'Engaged sessions per user', 'Events per session', 'Engagement rate', 'Event count', 'Hour'], dtype='object', name=0)

In [63]: `sns.set(style="whitegrid")`
`plt.figure(figsize=(10,5))`

```
df.groupby('DateHour')[["Sessions", 'Users']].sum().plot(ax=plt.gca())  
plt.title('Sessions and users overtime')  
plt.xlabel('DateHour')  
plt.ylabel('Count')  
plt.show()
```



The graph indicates consistent user activity with recurring traffic cycles. Sessions remain higher than users, showing repeated engagement. Peak traffic periods should be prioritized for business activities, while abnormal spikes should be further investigated for potential campaign impact.

Total Users by channel

```
In [64]: df.columns
```

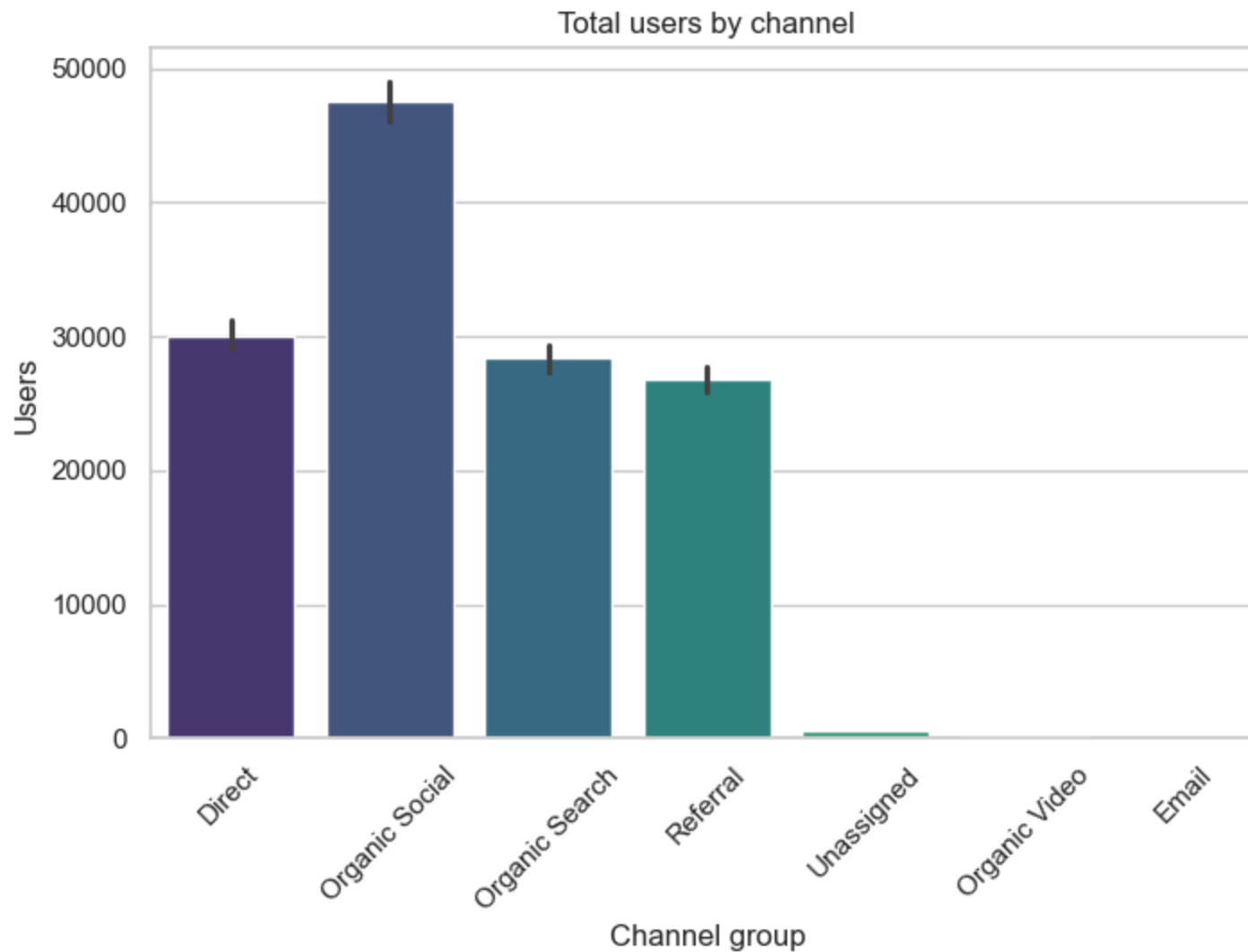
```
Out[64]: Index(['Channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions',  
              'Average engagement time per session', 'Engaged sessions per user',  
              'Events per session', 'Engagement rate', 'Event count', 'Hour'],  
              dtype='object', name=0)
```

```
In [65]: plt.figure(figsize=(8,5))  
sns.barplot(data=df,x="Channel group",y="Users",estimator=np.sum,palette='viridis')  
plt.title("Total users by channel")  
plt.xticks(rotation=45)  
plt.show()
```

C:\Users\abrar\AppData\Local\Temp\ipykernel_11880\3149503817.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=df,x="Channel group",y="Users",estimator=np.sum,palette='viridis')
```



From the above graph we can see that organic social is the highest performing channel indicating that strong social media engagement. Also a huge amount of users come in the website through direct way and organic search indicating that good brand awareness and SEO performance. So, investment should be taken in these channels. However email and organic video show very low engagement and may require strategic improvement.

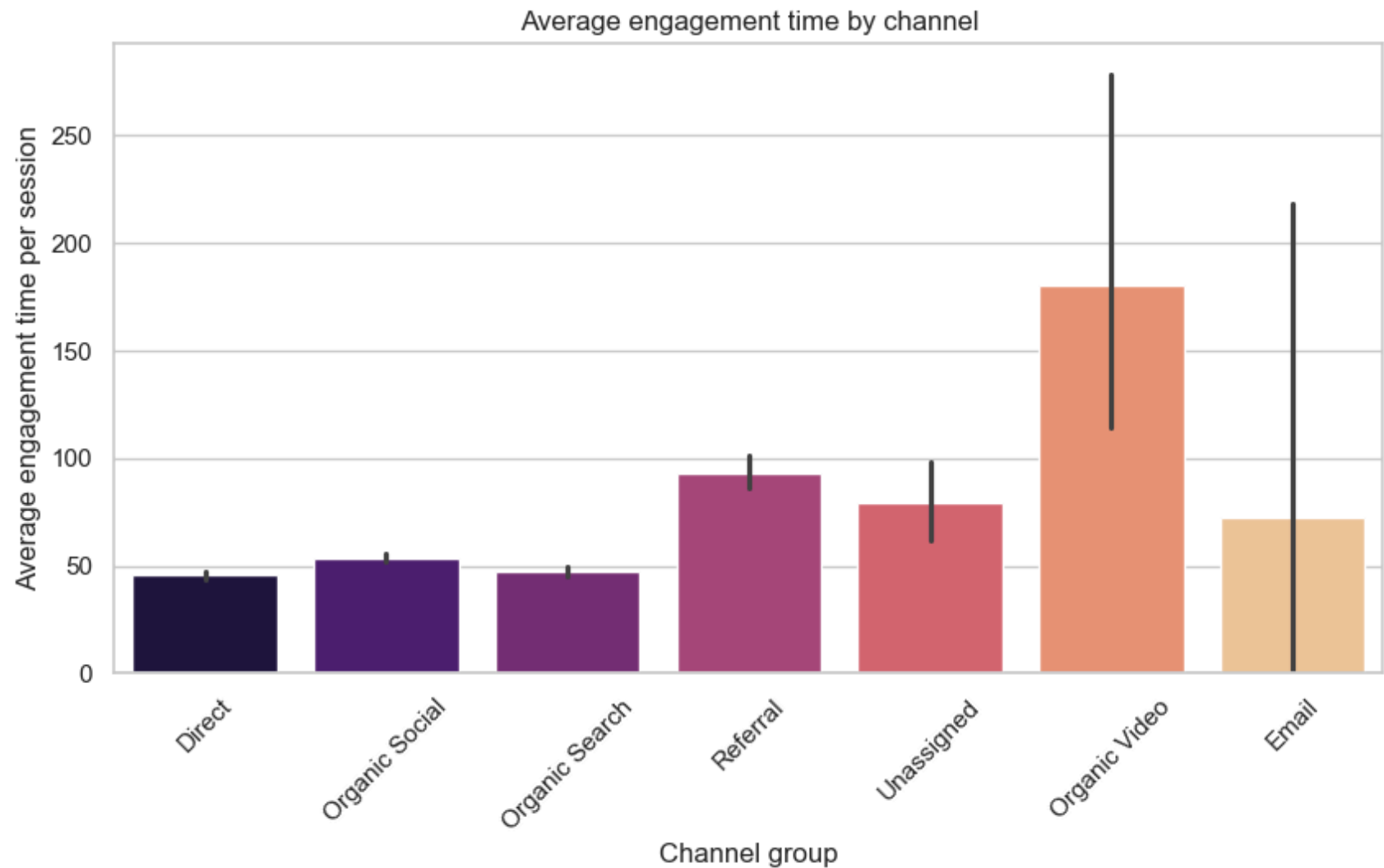
Average engagement time by channel

```
In [66]: plt.figure(figsize=(10,5))
sns.barplot(data=df,x="Channel group",y='Average engagement time per session',estimator=np.mean,palette='magma')
plt.title('Average engagement time by channel')
plt.xticks(rotation=45)
plt.show()
```

C:\Users\abrar\AppData\Local\Temp\ipykernel_11880\3856504018.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=df,x="Channel group",y='Average engagement time per session',estimator=np.mean,palette='magma')
```



From the graph we can see that organic video has the highest engagement time that means organic video provide high quality content that hold the traffic at a long time. So, necessary marketing strategy should be taken to bring more traffic through organic video.

Engagement rate distribution by channel

```
In [67]: df.columns
```

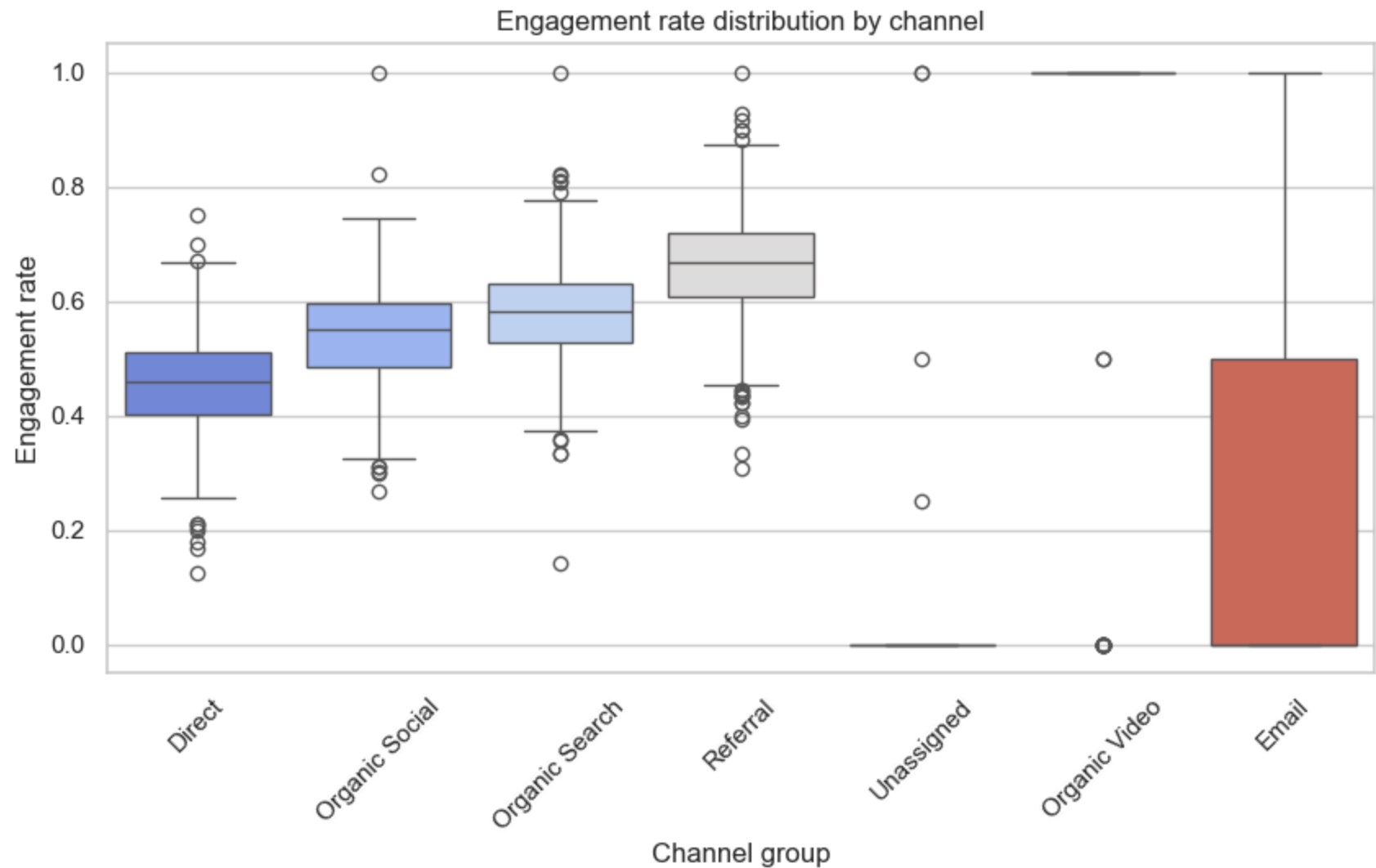
```
Out[67]: Index(['Channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions',  
              'Average engagement time per session', 'Engaged sessions per user',  
              'Events per session', 'Engagement rate', 'Event count', 'Hour'],  
              dtype='object', name=0)
```

```
In [68]: plt.figure(figsize=(10,5))  
sns.boxplot(data=df,x="Channel group",y='Engagement rate',palette='coolwarm')  
plt.title('Engagement rate distribution by channel')  
plt.xticks(rotation=45)  
plt.show()
```

C:\Users\abrar\AppData\Local\Temp\ipykernel_11880\1663733819.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df,x="Channel group",y='Engagement rate',palette='coolwarm')
```



Engagement Rate Distribution Analysis by Channel

This box plot was used to analyze the distribution, variability, stability, consistency, and outliers of engagement rates across different traffic channels.

Key Insights

- Referral channel shows the highest median engagement rate, indicating that referral users are generally more engaged than users from other channels.
- Organic Search and Organic Social channels show relatively stable and consistent engagement behavior because their engagement values are concentrated within a narrower range.
- Direct traffic has comparatively lower engagement, suggesting that direct visitors may interact less with the platform.
- Email channel shows very high variability because its engagement values range from very low to very high. This indicates inconsistent user behavior and unstable campaign performance.
- Organic Video and Unassigned channels contain extreme values (0 and 1), which may indicate low sample size, tracking issues, or data quality problems.

Variability Analysis

- Variability refers to how much the engagement values fluctuate.
- A larger box and longer whiskers indicate higher variability.
- Email channel has the highest variability, meaning user engagement changes significantly across users.
- Organic Search has lower variability, indicating more predictable engagement behavior.

Stability Analysis

- Stable data means the engagement values do not fluctuate heavily.
- Organic Search appears more stable because its engagement values remain within a controlled range.
- Stable channels are important because they provide more reliable and predictable performance for business decisions.
- Email channel appears unstable due to its large spread and inconsistent engagement values.

User Consistency

- Consistent users behave similarly and produce engagement values within a similar range.
- Organic Social users appear relatively consistent because most engagement values are clustered around the middle range.
- Email users appear inconsistent because some users are highly engaged while others show almost no engagement.

Outlier Analysis

- The circles outside the whiskers represent outliers.
- Outliers indicate unusually high or unusually low engagement behavior.
- Referral channel contains several high-engagement outliers, meaning some users interacted far more than typical users.

- Outliers may also indicate unusual behavior patterns or possible tracking anomalies.

Overall Conclusion

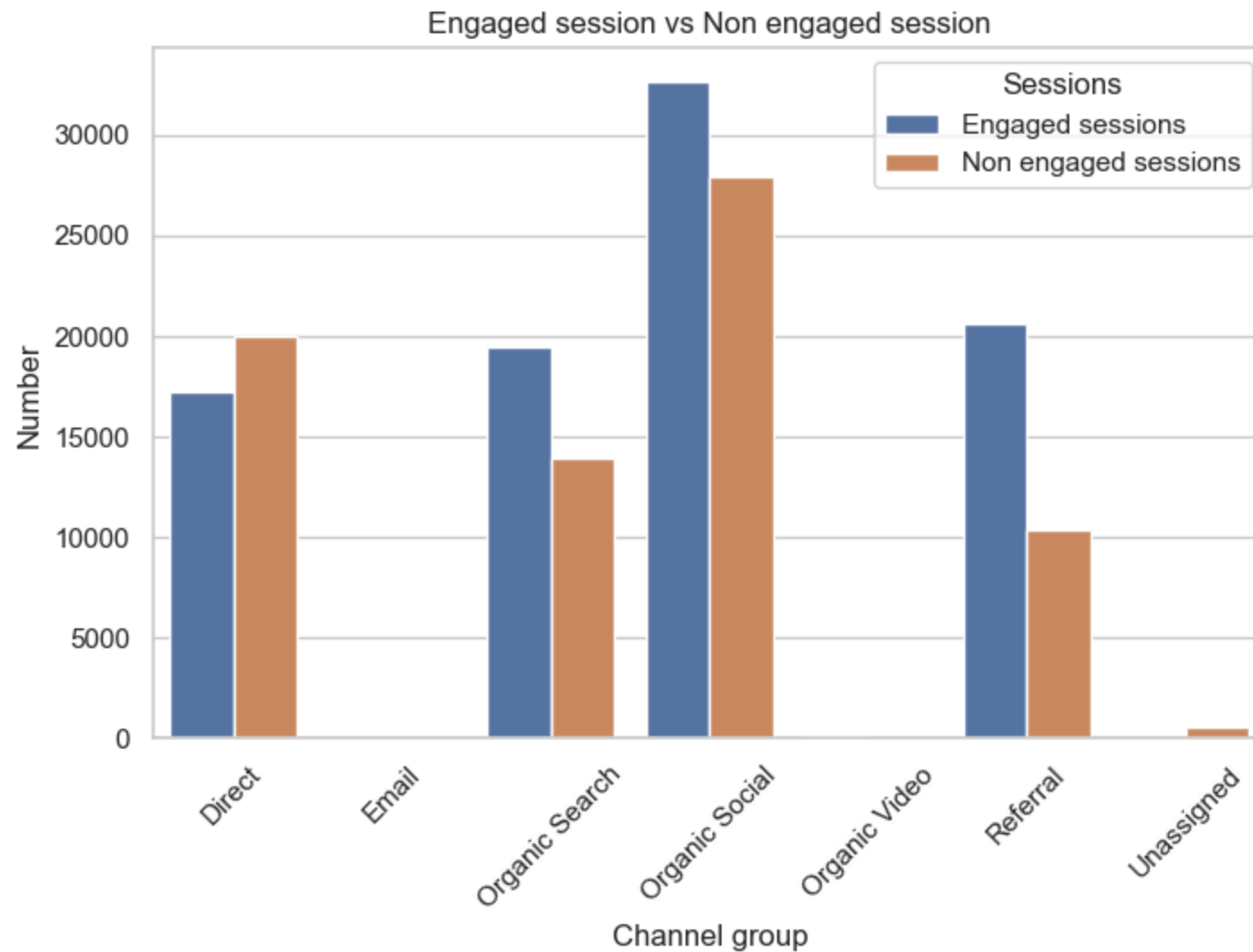
- Referral traffic provides the highest-quality engagement.
- Organic Search and Organic Social channels provide more stable and reliable engagement performance.
- Email campaigns may require optimization because user engagement is highly inconsistent.
- Some channels may need further data validation due to extreme values and outliers.

Engaged session vs Non engaged session

```
In [69]: df.columns
```

```
Out[69]: Index(['Channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions',  
              'Average engagement time per session', 'Engaged sessions per user',  
              'Events per session', 'Engagement rate', 'Event count', 'Hour'],  
              dtype='object', name=0)
```

```
In [70]: df_session= df.groupby('Channel group')[['Sessions','Engaged sessions']].sum().reset_index()  
df_session['Non engaged sessions']=df_session['Sessions']-df_session['Engaged sessions']  
df_session_melted=df_session.melt(id_vars='Channel group',value_vars=['Engaged sessions','Non engaged sessions'],var_  
df_session_melted  
plt.figure(figsize=(8,5))  
sns.barplot(data=df_session_melted,x='Channel group',y='Number',hue='Sessions')  
plt.title('Engaged session vs Non engaged session')  
plt.xticks(rotation=45)  
plt.show()
```



Insights from Engaged vs Non-Engaged Session Analysis

- Organic Social generated the highest traffic and engagement, showing strong social media performance.
- Organic Search also performed well, indicating effective SEO strategies.
- Referral traffic had significantly higher engaged sessions, suggesting high-quality external traffic sources.
- Direct traffic showed more non-engaged sessions, which may indicate poor landing page experience or weak user retention.
- Email traffic was extremely low, suggesting that email marketing is either inactive or ineffective.

- A small amount of unassigned traffic indicates possible issues in traffic source tracking or analytics configuration.

Business Decisions

- Continue investing in social media and SEO strategies.
- Improve user experience for direct visitors.
- Increase referral partnerships and backlink strategies.
- Develop stronger email marketing campaigns.
- Audit analytics tracking setup for proper traffic classification.

Traffic by hour and channel

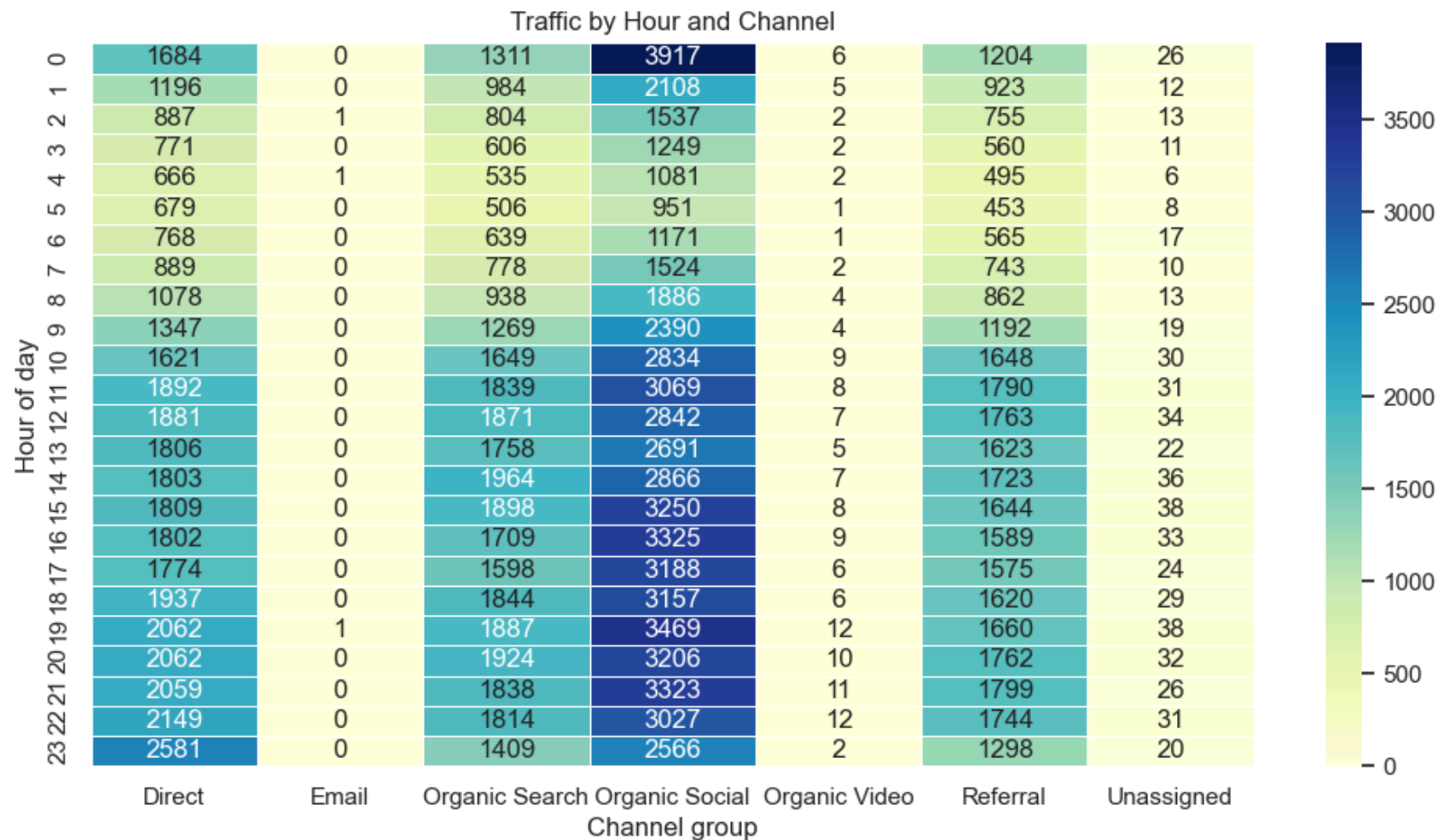
```
In [71]: heatmap_data=df.groupby(['Hour', 'Channel group'])['Sessions'].sum().unstack().fillna(0)
heatmap_data
```

Out[71]:

Channel group	Direct	Email	Organic Search	Organic Social	Organic Video	Referral	Unassigned
Hour							
0	1684.0	0.0	1311.0	3917.0	6.0	1204.0	26.0
1	1196.0	0.0	984.0	2108.0	5.0	923.0	12.0
2	887.0	1.0	804.0	1537.0	2.0	755.0	13.0
3	771.0	0.0	606.0	1249.0	2.0	560.0	11.0
4	666.0	1.0	535.0	1081.0	2.0	495.0	6.0
5	679.0	0.0	506.0	951.0	1.0	453.0	8.0
6	768.0	0.0	639.0	1171.0	1.0	565.0	17.0
7	889.0	0.0	778.0	1524.0	2.0	743.0	10.0
8	1078.0	0.0	938.0	1886.0	4.0	862.0	13.0
9	1347.0	0.0	1269.0	2390.0	4.0	1192.0	19.0
10	1621.0	0.0	1649.0	2834.0	9.0	1648.0	30.0
11	1892.0	0.0	1839.0	3069.0	8.0	1790.0	31.0
12	1881.0	0.0	1871.0	2842.0	7.0	1763.0	34.0
13	1806.0	0.0	1758.0	2691.0	5.0	1623.0	22.0
14	1803.0	0.0	1964.0	2866.0	7.0	1723.0	36.0
15	1809.0	0.0	1898.0	3250.0	8.0	1644.0	38.0
16	1802.0	0.0	1709.0	3325.0	9.0	1589.0	33.0
17	1774.0	0.0	1598.0	3188.0	6.0	1575.0	24.0
18	1937.0	0.0	1844.0	3157.0	6.0	1620.0	29.0
19	2062.0	1.0	1887.0	3469.0	12.0	1660.0	38.0
20	2062.0	0.0	1924.0	3206.0	10.0	1762.0	32.0

Channel group	Direct	Email	Organic Search	Organic Social	Organic Video	Referral	Unassigned
Hour							
21	2059.0	0.0	1838.0	3323.0	11.0	1799.0	26.0
22	2149.0	0.0	1814.0	3027.0	12.0	1744.0	31.0
23	2581.0	0.0	1409.0	2566.0	2.0	1298.0	20.0

```
In [72]: plt.figure(figsize=(12,6))
sns.heatmap(data=heatmap_data,cmap='YlGnBu',linewidths=.5,annot=True,fmt='.0f')
plt.title('Traffic by Hour and Channel')
plt.xlabel('Channel group')
plt.ylabel('Hour of day')
plt.show()
```



Traffic by Hour and Channel Analysis

The heatmap analysis reveals clear patterns in website traffic behavior across different hours and traffic channels.

Key Insights

- Organic Social was the strongest traffic source throughout the day and generated the highest overall traffic among all channels.
- The single highest traffic value was recorded at 12 AM (Hour 0) from Organic Social, indicating very strong late-night audience activity.

- Traffic generally increased after 9 AM and remained consistently high until late evening.
- The most consistently active period for overall website traffic was between 10 AM and 10 PM.
- Direct traffic became stronger during evening and late-night hours, suggesting the presence of returning or loyal visitors.
- Organic Search traffic remained stable throughout the day, showing consistent SEO performance.
- Referral traffic also contributed a significant amount of traffic during active hours, indicating valuable external traffic sources.
- Email and Organic Video channels generated very low traffic compared to other channels.
- A small amount of Unassigned traffic suggests possible analytics or traffic classification issues.

Business Decisions

- Focus marketing campaigns and important content publishing during afternoon and evening hours.
- Schedule social media posts and promotions during late-night hours, especially around 8 PM to 12 AM.
- Continue investing in social media marketing since Organic Social is the strongest traffic driver.
- Maintain and improve SEO strategies because Organic Search provides stable traffic throughout the day.
- Improve email marketing campaigns and video marketing strategies to increase engagement from weaker channels.
- Optimize website experience for late-night users, as audience activity is very high during nighttime.
- Audit analytics tracking and UTM configuration to reduce unassigned traffic sources.

Engagement rate vs Sessions over time

```
In [73]: df.columns
```

```
Out[73]: Index(['Channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions',  
              'Average engagement time per session', 'Engaged sessions per user',  
              'Events per session', 'Engagement rate', 'Event count', 'Hour'],  
             dtype='object', name=0)
```

```
In [74]: df_plot=df.groupby('DateHour')[['Engagement rate','Sessions']].mean().reset_index()  
df_plot
```


Out[74]:

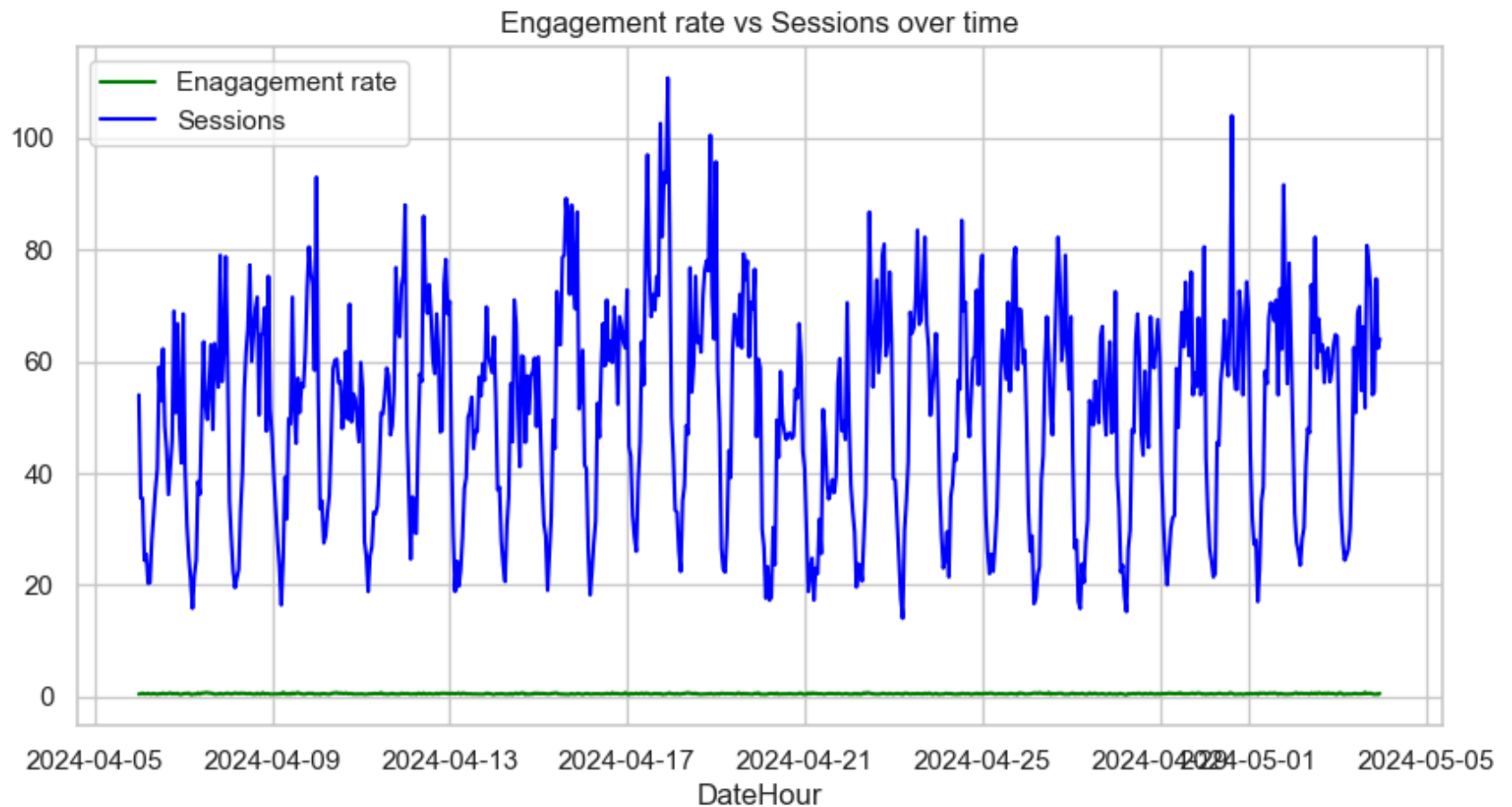
	DateHour	Engagement rate	Sessions
0	2024-04-06 00:00:00	0.450262	54.000000
1	2024-04-06 01:00:00	0.489905	35.500000
2	2024-04-06 02:00:00	0.595759	35.500000
3	2024-04-06 03:00:00	0.446725	24.400000
4	2024-04-06 04:00:00	0.494964	25.500000
...	...	...	...
667	2024-05-03 19:00:00	0.498125	54.000000
668	2024-05-03 20:00:00	0.375790	54.500000
669	2024-05-03 21:00:00	0.471312	74.800000
670	2024-05-03 22:00:00	0.367749	62.333333
671	2024-05-03 23:00:00	0.585665	64.000000

672 rows × 3 columns

```

In [75]: plt.figure(figsize=(10,5))
plt.plot(df_plot['DateHour'],df_plot['Engagement rate'],label='Enagagement rate',color='green')
plt.plot(df_plot['DateHour'],df_plot['Sessions'],label='Sessions',color='blue')
plt.title('Engagement rate vs Sessions over time')
plt.xlabel('DateHour')
plt.legend()
plt.grid(True)
plt.show()

```



Engagement Rate vs Sessions Over Time

This line graph compares the average **Sessions** and **Engagement Rate** over time based on the `DateHour` variable.

Key Insights:

- Website sessions fluctuate significantly over time, indicating varying user activity levels.
- Repeating peaks and drops suggest a periodic user behavior pattern, likely related to daily active and inactive hours.
- Several session spikes indicate periods of unusually high traffic, which may be caused by campaigns, promotions, or increased user interest.
- Overall traffic remains moderately consistent, with most session values staying within a common range.

- The engagement rate appears visually flat because it shares the same Y-axis with sessions, whose values are much larger.
- Due to the scale difference, the relationship between sessions and engagement rate cannot be analyzed clearly from this graph alone.